

Sharing Heartbeat: Toward Conducting Heartrate and Speech Rhythm through Tactile Presentation of Pseudo-heartbeats

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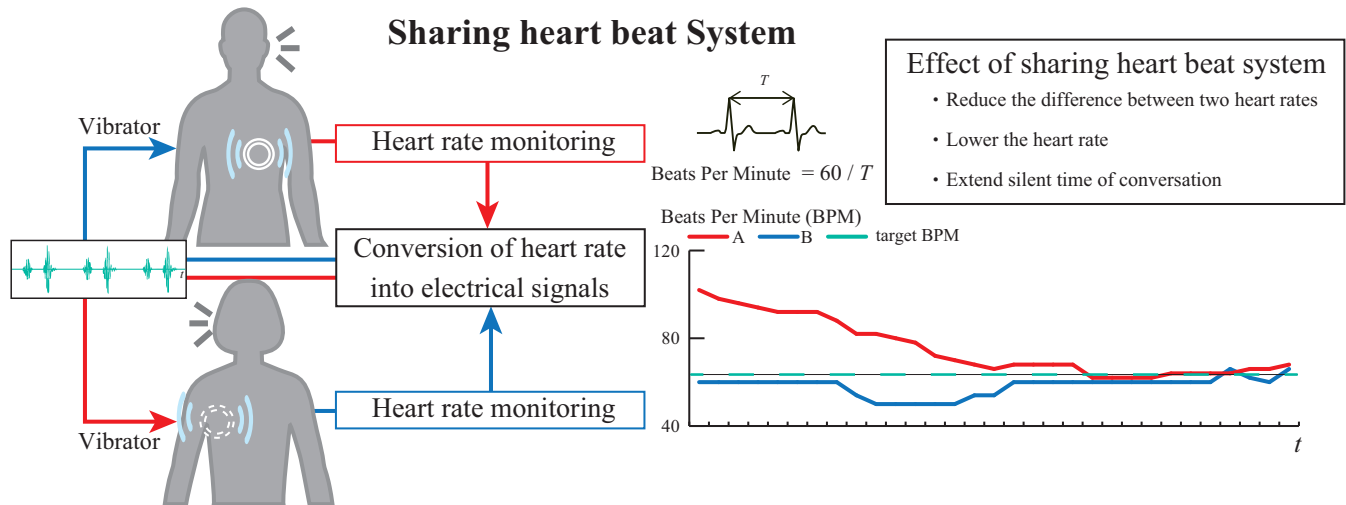


Figure 1: The device aims to achieve the same effect of physical contact lost with COVID-19 by sharing the heartbeat. The heartbeat is presented as a tactile sensation. The device conveys biometric information between two people and influences the actual heartbeat and speech rhythm.

ABSTRACT

Currently, the ongoing COVID-19 pandemic makes physical contact, such as handshakes, difficult. However, physical contact is effective in strengthening the bonds between people. In this study, we aim to compensate for the physical contact lost during the COVID-19 pandemic by presenting a pseudo-heartbeat through a speaker to reproduce entrainment and the synchronized state of heartbeats induced by physiological synchronization. We evaluated the effects of the device in terms of speech rhythm and heart

rate. The experimental results showed that a presentation of 80 BPM significantly reduced the difference in heart rate between the two participants, bringing them closer to a synchronized heart rate state. The heart rates of participants were significantly lower when 45 BPM and 80 BPM were presented than when no stimulus was given. Furthermore, when 45 BPM was presented, the silent periods between conversations were significantly more extended than when no stimulus was given. This result indicates that this device can intentionally create the entrainment phenomenon and a synchronized heart rate state, thereby producing the same effect of physical contact communication without contact.

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CCS CONCEPTS

• Human-centered computing → Interface design prototyping.

KEYWORDS

Entrainment, Physiological synchronization, Vibrotactile, Heart-beat

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1 INTRODUCTION

Since 2020, it has become necessary for people to keep a distance from each other to prevent infection due to COVID-19. The problem with communication in the current situation is that it has become difficult to engage in physical contact such as handshakes and hugs. Physical contact is effective in strengthening the bonds between people[9] and relieves stress and alleviate discomfort and distress[21].

Furthermore, physical contact has physiological effects on people. Heart rate variability and skin electrical activity tend to synchronize when people touch each other[10]. Such synchronization of physiological factors between individuals is called physiological synchronization[12]. Such synchronization of physiological factors helps empathy between people[1], strengthens social attachments among team members[36], and enables cooperation by building trust among team members[22].

As well as physical contact, conversation plays an essential role in building relationships between people. The tuning of various elements, both verbal[2] and non-verbal[3, 31], during conversation is called the chameleon effect[8], which creates a positive impression on the other party. Furthermore, speech rhythm also plays an important role in the conversation. Speech rhythm is determined by the length of the silent time (the time between one person's utterance and the other's) and the length of the vocal time (the time during which one person is speaking)[17]. In particular, the length of silent time is one of the most important elements of conversation because it indicates social connection[33], how well one anticipates the other's feelings[4, 28], and the degree of comprehension[5] of the content of the conversation.

Since biorhythmic synchronization and speech rhythm significantly impact mutual impressions, it is of great significance to intervene in these rhythms. In this study, we focused on entrainment[13, 14, 25]. Entrainment is a phenomenon in which two independent rhythms are influenced by each other. This phenomenon can be observed in gait[15, 32], biosignal[23, 29, 30], and cooperation with others[26, 27], indicating that stimuli easily influence human rhythm.

We focused on tactile stimuli. This is because physical contact is an essential part of communication, and it has been suggested that tactile stimuli presented through a device can have the same effect as actual contact[34]. Various studies have been conducted on social touch technology, which refers to tactile interfaces in the context of communication[16]. Further more, studies that manipulate the user's physiological state by presenting false heartbeat information

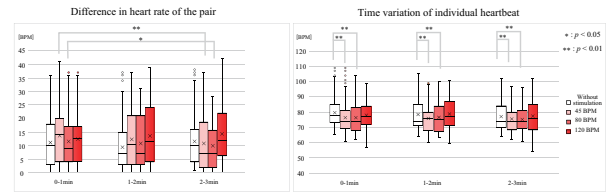


Figure 2: Box-and-whisker plots of the difference between the heart rates of the pairs (left panel) and individual heart rate changes over 3 minutes (right panel). There is no significant difference between the heart rate differences from 0 to 1 minute and 2 to 3 minutes in the case of no stimulus. However, there is a significant difference between 45 BPM and 80 BPM, indicating that there is an effect of reducing the difference between the heart rates of the two individuals. Individual heart rate changes over 3 minutes showed that 45 BPM and 80 BPM significantly decreased heart rate compared to no stimulation.

through visual and tactile senses have achieved specific results[12, 24, 35], and are thought to help induce entrainment. While these aim to reproduce human-to-human contact with a device[7, 11, 18], we focused on the heartbeat as the sense of touch presented.

In order to intentionally induce the entrainment, we presented a heartbeat, a typical biological rhythm, and a periodic stimulus from the device and shared it between two people during communication. The following hypotheses were formulated regarding the results.

- Hypothesis 1: It is possible to reduce the difference in heart rate between two people by tactile stimulation of a pseudo-heartbeat using a dynamic speaker on the chest.

The same BPM heartbeat can be given as a vibration to two communicating persons, reducing the difference in heart rate between the two persons. Suppose we can intentionally induce the same synchronization of heartbeats as the physiological synchronization produced by physical contact. In that case, tactile stimulation can realize the same effect as physical contact and improve the impression of each other's heartbeats.

- Hypothesis 2: Pseudo-heartbeat stimulation affects the heart rate and speech rhythm.

Based on the entrainment, it is considered that the heart rate and speech rhythm are affected by the pseudo-heartbeat stimuli. If the BPM affects the heart rate and speech rhythm, the heart rate and speech rhythm can be intentionally manipulated. This may lead to improvement of communication experience.

In this paper, the main contribution of our work is to intentionally create the same situation as the entrainment and physiological synchronization, aiming to compensate for physical contact lost in COVID-19 by sharing a pseudo-heartbeat using tactile stimuli and to intervene in biological rhythm and speech rhythm through tactile stimuli. We created a device that can share a heartbeat and conducted an experiment in which two pairs of people were asked to share a pseudo-heartbeat. We evaluated the results in terms of speech rhythm and heart rate.

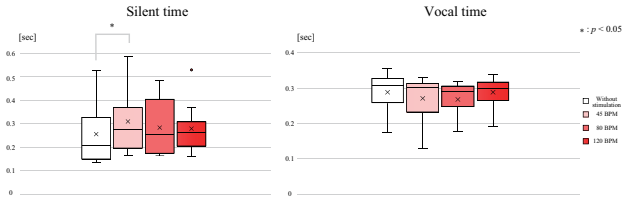


Figure 3: Box-and-whisker plots of silent time (left panel) and vocal time (right panel) in the alphabet task. There was a significant difference between 45 BPM and no stimulus in silent time, indicating an effect of lengthening the silent time. No significant difference existed in the vocal duration.

2 METHOD

2.1 System overview

Dynamic speaker (EAS65P118D, Panasonic) vibrates the chest of two talking people by presenting heartbeats of the same heart rate to each of them. The heartbeat sound from the speaker is cut off at frequencies above 1 kHz so that it is not audible, and only vibrations are presented. The speaker is connected to an amplifier (AX-501), which enhances the heartbeat sound from the PC.

2.2 Experiment

The experiment participants were 22 male and female participants without audiovisual impairments (glasses, etc., could be used). The experiment started with a loudspeaker fixed to the chest and a Fitbit worn on the wrist to measure the heart rate. Four types of pseudo-heartbeats are used in this experiment. The four simulated heartbeats are 120 BPM, 80 BPM, 45 BPM, and no stimulus, respectively. For each BPM, the experiment is divided into two parts. The order of stimuli is shuffled for each pair to eliminate the influence of the order. In both of the following two parts, the speech rhythm was analyzed by recording the content of the conversation.

In the first part, pairs are asked to engage in a 3-minute conversation about a given topic while presented with a pseudo-heartbeat. The topic of the conversation changes at each change in BPM. The topic of conversation could be about their high school/university, family, hobbies, or future. To shuffle the relationship between the BPM of the pseudo-heartbeats and the agenda, the order of the agenda was fixed as follows: high school and university, family, hobbies, and future.

In the second part, in order to examine the effect of the pseudo-heartbeat on speech rhythm, we conducted an experiment in which pairs of participants were asked to pronounce the letters of the alphabet from A to G alternately while a pseudo-heartbeat was presented, with reference to the study by Kawasaki et al[17]. An experiment was conducted in which pairs of participants were asked to pronounce the alphabet from A to G alternately while a pseudo-heartbeat was presented. The task was repeated four times without a break while the same BPM as that used in the first part was given. After this second part, the participants rested until their heart rate settled to 60 to 80 BPM, which is typical resting heart rate, before conducting the next BPM.

3 RESULTS AND DISCUSSION

3.1 Overview

The left panel of Fig. 2 shows the difference in the heart rates of the pairs in the first part of the experiment in 1-minute increments. Paired t-tests were conducted to see if there was a significant difference between the difference in heart rates from 0 to 1 minute and from 2 to 3 minutes for each stimulus. Significant differences existed for 45 BPM and 80 BPM (45 BPM: $p < 0.01$), (80 BPM: $p < 0.05$). Given that no significant difference exists without stimulation, the difference in heart rates of the pairs was significantly reduced when stimuli of 45 BPM and 80 BPM were given.

The right panel of Fig. 2 shows the individual heart rate in 1-minute increments for the first part of the experiment. There were significant differences ($p < 0.01$) from 0 to 1, 1 to 2, and 2 to 3 minutes for the no stimulus and 45 BPM stimulus, and for the no stimulus and 80 BPM stimulus, respectively. The heart rates of participants were significantly lower when 45 BPM and 80 BPM stimuli were applied than when no stimuli were applied.

Figure. 3 shows the results of part 2 of the experiment, the mean silent and vocal time of the alphabet task. There were significant difference ($p < 0.05$) between the no stimulus and the 45 BPM stimulus during the silent time. The results show that when the stimulus is 45 BPM, the silent time becomes longer and speech rhythm becomes slower.

3.2 Hypothesis 1: It is possible to reduce the difference in heart rate between two people by tactile stimulation of a pseudo-heartbeat using a dynamic speaker on the chest.

Heart rate variability is closely related to heart rate. Therefore, it can be said that a decrease in the difference in heart rate leads to a decrease in the difference in the value of heart rate variability. Since the difference in heart rate variability is also reduced when people make physical contact with each other[10], this result show that tactile stimulation by pseudo-heartbeat has the same effect as physical contact to a certain degree. Since the heart rate is closely related to the autonomic nervous system, it can be said that the degree of excitement and relaxation of two people become similar by the stimulation of pseudo-heartbeats.

3.3 Hypothesis 2: Pseudo-heartbeat stimulation affects the heart rate and speech rhythm.

Regarding the heart rate results, when the heart rate is lowered by stimulation of the pseudo-heartbeat, it indicates that the parasympathetic nervous system is dominant, and the participant can communicate more relaxed than when no stimulation is given.

Concerning the results of speech rhythm, people predict the timing of their next utterance during silent periods in conversation[6, 19, 20]. However, when the silent period is short, prediction of the timing of utterance is said to hinder comprehension of the conversational content[5]. Therefore, it can be said that slowing down the speech rhythm by using this device will promote comprehension of the conversational content.

These two considerations for two hypotheses indicate that this device can intentionally create the entrainment phenomenon and

synchronized heart rate state, therefore producing the effect of physical contact without contact.

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